Findings — reproduction and re-evaluation of *Emotion Unlocked*

Paper: Vasuki, Padhma Priya M, Pooja V — *Emotion Unlocked: Multimodal Emotion Detection Through Audio-Video Fusion*, SSN College of Engineering (preprint, SSRN 5274911). **Branch:** `actor-independent-eval` · **Date:** 2026-07-17

Everything below is measured, not asserted. Every claim names the cell, section, or run it comes from. Where a claim did not survive checking, it is marked as withdrawn rather than deleted.

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1. The published benchmark measures actor identity

The paper reports **96.06%** fusion accuracy (Table VII). Under actor-independent evaluation — the same architecture, the same data, but with the people in the test fold never seen during training — it scores far less:

model	actor-independent	sd	vs paper's 96.06
audio	**55.08**	3.54	-40.98
video	**45.43**	12.56	-50.63
fusion (paper's fusion protocol)	**62.98**	7.96	-33.08
fusion (out-of-fold features)	**50.25**	9.10	-45.81

Holding out actors costs **33.08** points. The fusion-feature leak costs a further **12.73**, measured on identical held-out actors, so the two are separable and additive.

This is a flaw in the publication, not in anyone's implementation. The paper's §F says:

> "five-fold cross-validation is adopted. The database is split into five subsets wherein: Four subsets are used for training. One subset is used for validation."

Subsets **of samples**. Actors and speakers are never mentioned anywhere in the paper. RAVDESS is 24 actors delivering **two fixed sentences**, each recorded **twice** — so repetition 1 lands in training while repetition 2 lands in validation: same face, same voice, same sentence, same emotion, same intensity. A model can score well by recognising *who is speaking*.

Corroborating signal, before any of our changes: the **video-only** model scores 91–93% under the paper's protocol. Video-only emotion recognition at 92.72% from 3-second clips is not plausible from facial expression; it is the memorisation signature showing directly.

Two leaks, and they are additive. The 33-point gap is actor leakage. A further **12.73** points comes from a second, independent leak: the paper builds the fusion MLP's *training* features by running the training clips through the models that trained on them (notebook cell 35), so the 8-dim softmax inputs are near-perfect one-hots in training and merely ~80% right at inference. The MLP learns to trust a signal that degrades the moment it is deployed. Measured on *identical* held-out actors, so it is isolated from the actor leak.

The model does read something. Errors cluster affectively — sad↔fearful (87), angry↔disgust (82), happy↔calm (79) — not randomly. The architecture works. It is just far weaker than advertised.

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2. The notebook never reproduced the paper

	audio	video	fusion
paper, Table VII	79.24	92.72	**96.06**

`Paper_worK/paper-notebook.ipynb`, as committed	~80.7	91.2	**94.29**

It lands **1.77 short**, and three deviations explain why it is not implementing the paper:

a) It trains on the wrong data — and the extra data is the *easy* half. Paper §VI.6: *"The dataset has 1,360 distinct samples, each with both an audio and video recording."* The notebook pairs **2,452**. Verified against the Kaggle file listing: RAVDESS gives 1,440 speech (24 actors × 60 trials) + 1,012 song (23 × 44 — actor 18 recorded no song). **Neither 2,452 nor 1,440 equals 1,360**, so the paper's own accounting does not reconcile with the dataset's structure.

The notebook's extra ~1,000 clips are the **song** subset, where actors *sing* rather than speak and where **disgust and surprised do not exist at all**. Measured on the held-out actor predictions, song is far *easier*:

model	speech	song	
audio	45.14	**69.27**	song **+24.13**
fusion	43.19	**59.78**	song **+16.59**
video	46.74	42.69	speech +4.05

Neutral alone runs 25.0% on speech against **84.8%** on song. Singing exaggerates emotional prosody — pitch, duration and intensity all become more distinct — which is why *audio* gains 24 points while video is unaffected.

So the notebook is, if anything, **flattered** by training on ~80% more data than the paper describes, most of it easier — and it *still* lands 1.77 points below the paper's number.

(We predicted the opposite and were wrong; see §5.)

b) It does not implement Fig. 4. Paper §VII.2 specifies r3d_18 → *"additional 3D convolutional layers are stacked, followed by global average pooling"*. The notebook slices `nn.Sequential(*list(backbone.children())[:-1])`, which **keeps r3d_18's own AdaptiveAvgPool3d**. The volume is therefore already 1×1×1 before the `Conv3d(512,256,k=3,padding=1)` stack runs, so each conv sees 26/27 zero-padding and degenerates to its centre tap — a stack of Linear layers wearing Conv3d costumes. Cell 21's own comment shows the author knew.

c) Its fused features are unscaled. Cell 33 concatenates raw MFCC means (≈ –300) with 0–1 softmax probabilities, and `FusionMLP`'s first `BatchNorm` sits *after* the first `Linear`, so nothing ever normalises the input.

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3. What we changed, and what it cost or bought

All arms below run the **paper's own protocol** — its `StratifiedKFold(shuffle=True)` split, its max-over-epochs metric, its hyperparameters — so every row is directly comparable to Table VII.

arm	what changed	audio	video	fusion	vs paper
paper, Table VII	*the bar*	79.24	92.72	**96.06**	—
notebook, as committed	*never reproduced its own paper*	80.67	83.24	94.29	–1.77
kinetics	Kinetics channel stats + scaler	81.81	81.97	**93.84**	-2.22
per-clip	per-clip norm + scaler	81.73	78.25	**94.25**	-1.81
+ Fig. 4	per-clip + scaler + refine head on a real 2x7x7 volume	81.16	92.62	**95.84**	-0.22

Per-fold detail:

kinetics | fold | audio | video | fusion | |---|---|---|---| | 1 | 78.21 | 85.54 | 94.70 | | 2 | 82.28 | 88.80 | 94.91 | | 3 | 84.69 | 71.22 | 93.27 | | 4 | 80.20 | 88.78 | 91.22 | | 5 | 83.67 | 75.51 | 95.10 | | **mean** | 81.81 | 81.97 | 93.84 | | *sd* | 2.62 | 8.11 | 1.63 |

per-clip | fold | audio | video | fusion | |---|---|---|---| | 1 | 80.45 | 82.48 | 94.70 | | 2 | 81.06 | 93.48 | 96.74 | | 3 | 84.69 | 79.39 | 95.71 | | 4 | 79.80 | 70.20 | 92.45 | | 5 | 82.65 | 65.71 | 91.63 | | **mean** | 81.73 | 78.25 | 94.25 | | *sd* | 1.96 | 10.88 | 2.16 |

+ Fig. 4 | fold | audio | video | fusion | |---|---|---|---| | 1 | 80.65 | 93.08 | 94.50 | | 2 | 80.65 | 92.06 | 97.35 | | 3 | 83.88 | 94.49 | 97.55 | | 4 | 74.08 | 91.43 | 94.08 | | 5 | 86.53 | 92.04 | 95.71 | | **mean** | 81.16 | 92.62 | 95.84 | | *sd* | 4.66 | 1.20 | 1.59 |

The finding: one line of the notebook costs 9.4 points of video

The paper's video number is exactly reproducible — but only if you build Fig. 4 as written.

	video mean	sd	fold range
paper, Table VII	**92.68**	3.05	88.2 – 96.6
ours, Fig. 4 implemented	**92.62**	**1.20**	91.4 – 94.5
notebook (pools before the convs)	83.24	—	(fold 1 = 91.24, its best)
kinetics (pools)	81.97	8.11	71.2 – 88.8
per-clip (pools)	78.25	10.88	65.7 – 93.5

92.62 against 92.72. A 0.10 gap, at *lower* variance than the published result (1.20 vs 3.05). Paired against the identical pipeline with the pooled head: **+14.37, t = +3.01, p = 0.039.**

The cause is a single slice. Paper §VII.2 specifies r3d_18 → *"additional 3D convolutional layers are stacked, followed by global average pooling"*. The notebook writes:

```
self.backbone = nn.Sequential(*list(backbone.children())[:-1])   # keeps r3d_18's OWN avgpool
```

`[:-1]` drops only the final `fc`, **keeping r3d_18's `AdaptiveAvgPool3d`**. The volume is therefore already 1×1×1 when the `Conv3d(512,256,k=3,padding=1)` stack runs — every conv sees 26/27 zero-padding and collapses to its centre tap. The refine head becomes three Linear layers wearing Conv3d costumes, and the paper's stated architecture is never built.

`[:-2]` keeps `stem..layer4`, emitting **(B, 512, 2, 7, 7)**, and the convs do the spatiotemporal work the paper describes. That one character is the entire 9.4-point video deficit — and it is essentially free (~0.35 GMAC/sample against r3d_18's own ~40 GFLOP).

This vindicates the paper and indicts the notebook. The published video result is real and reproducible; this repo's implementation simply is not the paper's architecture.

Our audio also **matches and exceeds** the paper throughout (81.2–81.8 vs 79.24), so no part of the pipeline is broadly broken.

Retracted: "Kinetics normalization is a 10.75-point regression"

This claim was wrong and is withdrawn. It came from comparing our video against the *paper's* 92.72 and against the notebook's *fold 1* (91.24 — its single best fold), rather than against the notebook's actual mean of **83.24**.

Measured correctly, against the notebook's mean:

arm	video vs notebook	fusion vs notebook
kinetics	–1.27	–0.45
per-clip	–4.99	–0.04

And a **paired t-test across identical folds** (same seed, same splits) says the two arms are indistinguishable:

```
video   perclip − kinetics = −3.72   t = −0.77   p = 0.486   not significant
fusion  perclip − kinetics = +0.41   t = +0.39   p = 0.719   not significant
audio   perclip − kinetics = −0.08   t = −0.13   p = 0.903   not significant
```

The normalization choice does not measurably matter here. With video sd of 8–11 across only 5 folds, the standard error is ~4–5 points — large enough to swallow both changes whole. An entire narrative was built on noise; it is retained here as §5 material rather than deleted.

What the StandardScaler is worth: nothing measurable

Fusion: notebook 94.29, per-clip+scaler **94.25** (–0.04), kinetics+scaler 93.84 (–0.45). The scaler was the one change with a clean mechanical rationale — raw MFCC means near –300 beside 0–1 softmax probabilities, with `FusionMLP`'s first BatchNorm sitting *after* the first Linear — and it moves nothing. The fusion MLP evidently copes with the scale mismatch on its own.

Fusion is also strikingly insensitive to video quality: video ranging 78.25 → 81.97 across arms moves fusion by 0.41. The audio branch and the softmax probabilities appear to carry the fusion result, with the 512-dim video embedding contributing little.

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4. The depressed / not-depressed head

RAVDESS contains no depression label. It is 24 actors portraying 8 emotions on two fixed sentences. `sad → depressed` builds a detector for *acted sadness*, which is not clinical depression in either direction: acted sadness is not depression, and depression frequently presents as blunted, flat affect rather than sad affect. Depression is measured over weeks (PHQ-8, BDI), not in a 3-second clip.

The measurement says the same thing, without needing the argument. Post-hoc over the saved out-of-fold predictions:

mapping	positive	accuracy	majority baseline	beats baseline by	AUC
`sad → depressed`	15.3%	**86.46**	84.67	**+1.79**	81.58
`sad + fearful`	30.7%	80.71	69.33	+11.38	83.27
`negative_valence`	53.8%	**79.12**	53.83	**+25.29**	86.68

The depression mapping reports the highest accuracy and learns the least. Because only 15.3% of RAVDESS is sad, always answering "not depressed" scores **84.67%** — the model beats a hardcoded constant by 1.79 points. `negative_valence` reports a *lower* number while doing **fourteen times** more work. On **audio alone** the depression mapping scores **84.38% against an 84.67% baseline** — *below* a constant classifier.

The valence head is real, not a class prior: per-emotion firing rates are angry .90, fearful .90, disgust .88, sad .74 against neutral .15, calm .17, happy .27. (*Surprised* at .64 is the honest wart — its valence is genuinely ambiguous.)

Recommendation: report `negative_valence` as the headline; keep the `depressed_*` rows beside it **with their baselines in the same table**. A reader who sees 86.46 next to 84.67 cannot be misled. A reader who sees 86.46 alone will be.

Cut for the deadline: fine-tuning on DAIC-WOZ's real PHQ-8 labels. It needs a signed access agreement with days of lead time. The paper should report the transfer pathway as demonstrated and clinical validation as pending data access.

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5. Claims we withdrew

Kept visible so they are not cited from earlier drafts.

- **"The video key collision is an accuracy bug."** The collision is real and total — every one of 2,452 video keys matches 2 files, confirmed against the Kaggle listing, because cell 7's `key` omits `modality`. But modality 01 (full-AV) and 02 (video-only) are the *same visual recording*, so `v.iloc[0]` picking either yields identical frames. **Cosmetic**: it buys determinism, not accuracy.

- **"Kinetics normalization is a fix."** Withdrawn — no measurable effect either way (§3).
- **"Kinetics normalization is a 10.75-point regression."** Also withdrawn, and the more instructive error. Having found the first claim wrong, I over-corrected into an equally confident claim in the opposite direction — built from comparing against the paper's 92.72 and the notebook's single best fold rather than its mean. A paired t-test says p=0.486. **Being wrong twice in opposite directions about the same 20 lines of code is the signature of reading noise as signal**, and the fix was not a better theory but a baseline (the notebook's actual mean) and a significance test.
- **"The paper's video number is not reproducible."** Withdrawn within the hour. It reproduces to **0.10** once Fig. 4 is actually implemented (§3). The claim was made while every arm on the bench happened to share the notebook's architecture bug — an absence of evidence read as evidence of absence. The correct statement was always the narrower one: *no reproduction that pools before the refine convs reaches it.*
- **"Song is a contaminating domain; dropping it will raise accuracy."** Backwards. Song is the *easier* subset by 16–24 points (§2a). Dropping it would delete the easy half and lower the score. The arm was built and then cancelled before it consumed a GPU slot, on the strength of the saved predictions.
- **"The `refine` head is a bug."** It is a deliberate, documented deviation (cell 21's comment says so) — but it *does* mean the notebook doesn't implement the paper's Fig. 4. Reclassified from bug to unimplemented specification.
- **"The target is 94.29%."** That is the notebook's score. The paper's is 96.06%.

6. Known limitations

- **The metric is the paper's**: max-over-epochs accuracy on the very fold being reported. Optimistic by construction. Kept deliberately — changing it would make comparison to Table VII meaningless. You cannot claim to beat a number you measured differently.
- **Out-of-fold features use inner 2-fold**, not full nested CV — the affordable approximation. Training features come from models trained on half the training actors; test features from models retrained on all of them (standard stacking, as `sklearn`'s `StackingClassifier` does).
- **The paper does not state its epoch counts.** The notebook uses 15 for video, and its own fold-1 log shows validation accuracy still rising at epoch 15 (90.02 → 91.24). Video is plausibly under-trained, and the paper's max-over-epochs metric rewards longer training — so a fair comparison would need an epoch count the paper never published.
- **The actor-independent numbers in §1 are a lower bound**: they were produced with the Kinetics regression in place. The per-clip re-run will raise them.